

Lecture Notes: Risk, Return, CAPM, and Bond Valuation

Corporate Finance
Chapters 6, 7, 13

I. Systematic vs. Unsystematic Risk

Risk Decomposition

The total surprise element in returns can be broken into two parts:

$$\text{Actual Return} = E(R) + \text{Systematic Portion} + \text{Unsystematic Portion}$$

Systematic Risk (Market Risk)	Unsystematic Risk (Unique/Firm-Specific)
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Affects many assets simultaneously	Affects specific companies or assets
Sources: GDP, interest rates, inflation, wars	Sources: product failure, labor strikes, leadership changes
Cannot be eliminated through diversification	Can be eliminated through diversification
Also called non-diversifiable risk	Also called diversifiable risk

II. The Principle of Diversification

Key Insight

As the number of assets in a portfolio increases, **unsystematic risk decreases** and approaches zero. Only systematic risk remains.

Why Diversification Works

- When one stock goes up and another goes down, they balance each other out.
- In a large portfolio (e.g., all KSE 100 Index stocks), positive and negative firm-specific events cancel out.
- The net effect of unsystematic risk becomes relatively small.

Total Risk Decomposition

$$\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}$$

- Unsystematic risk: Can be reduced or eliminated
- Systematic risk: Cannot be avoided; the market only rewards you for this risk

III. Measuring Systematic Risk: Beta (β)

The **systematic risk principle** states that the return of a risky asset depends only on its systematic risk. The market does not reward you for risk that can be avoided through diversification.

Beta Coefficient

Beta measures the sensitivity of a firm's risk relative to the overall market.

Beta Value	Interpretation
$\beta = 1.0$	Average asset; moves with the market
$\beta > 1.0$	Higher systematic risk (e.g., tech stocks, Shopify)
$\beta < 1.0$	Lower systematic risk (e.g., Coca-Cola, McDonald's, J&J)

Beta vs. Standard Deviation

- **Standard Deviation:** Measures **total risk** (systematic + unsystematic)
- **Beta:** Measures **only systematic risk**

Example: Comparing Two Securities

Security	Standard Deviation	Beta
Security A	40% (high total risk)	0.50 (low systematic risk)
Security B	20% (lower total risk)	1.50 (higher systematic risk)

Conclusion: Required return is determined by **Beta**, not total risk, because unsystematic risk can be diversified away.

Portfolio Beta

Portfolio Beta is the weighted average of individual asset betas:

$$\beta_p = \sum_{i=1}^n w_i \beta_i$$

Example 1: Two-Stock Portfolio

- Shopify: $\beta = 1.59$, weight = 50%
- Coca-Cola: $\beta = 0.55$, weight = 50%

$$\beta_p = (0.5 \times 1.59) + (0.5 \times 0.55) = 0.795 + 0.275 = 1.07$$

Example 2: Unequal Weights

- Stock 1: $\beta = 0.8$, investment = \$35,000
- Stock 2: $\beta = 1.4$, investment = \$40,000
- Total investment = \$75,000

$$w_1 = \frac{35,000}{75,000} = 0.467 \quad w_2 = \frac{40,000}{75,000} = 0.533$$

$$\beta_p = (0.467 \times 0.8) + (0.533 \times 1.4) = 0.3736 + 0.7462 = 1.12$$

IV. Capital Asset Pricing Model (CAPM)

The CAPM defines the required return an investor demands for a stock based on its systematic risk.

CAPM Equation

$$r = r_{RF} + \beta \times (r_m - r_{RF})$$

Where:

- r = Required return on the stock
- r_{RF} = Risk-free rate
- r_m = Expected market return
- $(r_m - r_{RF})$ = Market risk premium
- β = Systematic risk measure

Example: CAPM Calculation

- $r_{RF} = 5\%$
- Market risk premium = 6% (so $r_m = 11\%$)
- $\beta = 1.2$

$$r = 5\% + 1.2 \times (11\% - 5\%) = 5\% + 1.2 \times 6\% = 5\% + 7.2\% = 12.2\%$$

Large Fund Case Study

A money manager has a \$4 million fund with four stocks:

Stock	Investment	Weight	Beta
A	\$400,000	10%	?
B	\$600,000	15%	-0.5
C	\$1,000,000	25%	?
D	\$2,000,000	50%	?

Step 1: Calculate Portfolio Beta = 0.7625

Step 2: Calculate Required Return (given $r_{RF} = 4\%$, $r_m = 8\%$)

$$r_{fund} = 4\% + 0.7625 \times (8\% - 4\%) = 4\% + 0.7625 \times 4\% = 4\% + 3.05\% = 7.05\%$$

V. Bond Valuation: YTM Approximation Formula

Calculating exact YTM requires trial and error. The approximation formula provides a close estimate:

YTM Approximation Formula

$$YTM \approx \frac{C + \frac{FV - P}{n}}{\frac{FV + P}{2}}$$

Where:

- C = Annual coupon payment
- FV = Face value (par value)
- P = Current price
- n = Years to maturity

Example

- Face value = \$1,000
- Coupon rate = 10% ($C = \$100$)
- Price = \$930
- Years to maturity = 8

$$\text{Numerator} = 100 + \frac{1000 - 930}{8} = 100 + \frac{70}{8} = 100 + 8.75 = 108.75$$

$$\text{Denominator} = \frac{1000 + 930}{2} = \frac{1930}{2} = 965$$

$$\text{Approximate YTM} = \frac{108.75}{965} = 11.26\%$$

$$\text{Exact YTM (calculator)} = 11.37\%$$

VI. Important Bond Features for Midterm

Bond Type	Description
Convertible Bond	Can be exchanged for common stock at the holder's option
Warrant	Long-term option to buy shares at a specified price
Callable Bond	Issuer can retire the bond before maturity
Puttable Bond	Holder can sell the bond back to the company prior to maturity
Bearer Bond	Physical coupons; no ownership record
Registered/Digital Bond	Ownership tracked in a database

VII. Midterm 2 Important Topics

Midterm 2 Syllabus - Important Topics

Chapter 5: Time Value of Money

- **Amortization Tables:** Be able to construct and calculate principal and interest components

Chapter 6: Interest Rates

- Interest rate components: $r = r^* + IP + DRP + LP + MRP$
- Yield curve construction and interpretation
- Pure Expectations Theory and forward rate calculations

Chapter 7: Bonds and Valuation

- Bond valuation (PV of coupons + PV of face value)
- Premium, par, and discount bonds
- YTM calculation (exact and approximation formula)
- Bond features: convertible, warrant, callable, putable, bearer vs. registered

Chapter 13: Risk and Return

- Expected return calculation
- Variance and standard deviation
- Coefficient of Variation
- Portfolio expected return
- Systematic vs. Unsystematic risk
- Beta coefficient and portfolio Beta
- **CAPM equation:** $r = r_{RF} + \beta(r_m - r_{RF})$

Exam Tips

- The midterm exam is lengthy; manage your time well.
- Practice calculations for expected returns, standard deviations, and CAPM.
- Know the difference between bond types and their features.

Summary of Key Formulas

Portfolio Beta: $\beta_p = \sum w_i \beta_i$

CAPM: $r = r_{RF} + \beta(r_m - r_{RF})$

YTM Approximation: $YTM \approx \frac{C + (FV - P)/n}{(FV + P)/2}$

Total Return: $R = E(R) + \text{Systematic} + \text{Unsystematic}$